M.A.C.H. Project Brief

Multimodal Agentic Copilot for the sHop floor: a trust-first assistant for SOPs and work instructions · TECHgium 10th edition, presentation round · Prepared 5 November 2026 for Sachiv

12 Nov, 11:59 PM IST
PPT deadline, no extensions

15 slides max
Use the LTTS template; rules on slide 13

7 + 5 min
Presentation, then jury Q&A, on Microsoft Teams

7 days
Build a thin working slice, measure it, pitch it

The pitch in one line: M.A.C.H. is a shop-floor assistant that answers from the plant's own SOPs, diagrams and tables, cites the exact step it used, refuses or calls an expert when unsure, and never answers from a superseded revision.

Why it can win. Image-based document search and self-checking RAG are already published at ICLR and NAACL, so on their own they are table stakes. What we found *no peer-reviewed work on* is the trust layer for safety-critical procedures: step-level revision safety (outdated documents in general are studied, procedures are not), step-level safety-complete citations, hazard-tiered abstention, and AI drafts measured under change control. That layer is M.A.C.H.'s novelty. Every claim below is worded "we found no prior work", not "world first", because IEEE and Elsevier full texts were checked through the OpenAlex index, not read in full.

1. The problem

Operators follow SOPs and work instructions (WIs) that change through revisions, engineering change notices (ECNs) and temporary deviations. Key values (torques, temperatures, sequences) often live in tables and diagrams, not text. Keyword search misses those. Plain chatbots and plain RAG add a new risk: a fluent, confident, wrong answer.

- **Outdated documents flip answers.** One stale document overturns 17–91% of otherwise-correct RAG answers arXiv 2026, not peer-reviewed. Date-based re-ranking recovers only 4.6–10 points; stating explicitly *when* old evidence stops applying fixes most of it. Peer-reviewed work confirms the harm HoH, ACL 2025 and shows version-unaware evaluation misreports recall by up to 25.7 points EMNLP 2026 Industry, but neither covers procedures.
- **Citations are not proof.** Commercial legal RAG tools hallucinate on 17–33% of queries, including citations that don't support the claim J. Empirical Legal Studies 2025. In medicine, 50–90% of LLM answers are not fully supported by their own sources Nature Communications 2025.
- **Industry has no benchmark.** The peer-reviewed manufacturing RAG papers we found evaluate retrieval only Procedia CIRP 2025. Nobody has published a hallucination audit of generative RAG on SOPs.
- **Auditors care.** ISO 9001 §7.5 requires that only the current revision is available at the point of use; an obsolete WI at a station is a common nonconformity. 21 CFR 211.100(b) requires every deviation to be recorded and justified.

2. What M.A.C.H. delivers

Challenge deliverable	How M.A.C.H. meets it	Novel part (N)
Versioned multimodal corpus with access control	Revision registry with status, effective dates, scope (line, model, serial range), supersedes links and deviation overlays. Role filter applied before search.	N Step-level deviation overlays with expiry
Ingest text, tables, images	Layout parser gives every step, table and figure a stable ID; page images kept for visual search; scanned pages routed to image search.	Cross-reference links ("see Fig. 3, step 4")

Multimodal answers with citations and confidence	Four search lanes fused, then an answer where every step cites doc, revision, page, step and figure ID. Numbers copied from source cells by code.	**N** Step-level safety-complete citations; number lock; context lock
Abstention, conflicts, freshness, escalation	Evidence grader, conflict detector, Revision Guard, hazard-tiered confidence gate, partial answers, expert inbox with candidate answers.	**N** Revision Guard; hazard-tiered and shift-aware abstention
Change-controlled WI drafting	Drafter and Compliance agents produce a draft from an RCA; human approval required; approved draft becomes a new revision and the old one is superseded.	**N** Planted-fault reviewer test; STE rule check
Evaluation vs keyword search and text-only RAG	MACH-Bench (about 250 questions in 8 buckets, built from real FAA Airworthiness Directive supersessions plus scripted counterfactual edits) plus 30 drafting tasks, and eight public benchmarks; paired statistics; ablations.	**N** First multimodal SOP benchmark with revision traps
Security and deployment report	RBAC, audit log with replay, prompt-injection defences, cloud and on-prem paths, keyword-only fallback when the LLM is down.	Audit replay of any past answer

3. System design

M.A.C.H. has three parts. Offline ingestion turns documents into searchable, revision-tagged pieces. The online answer path runs every question through a chain of checks before anything reaches the operator. Governance handles experts, change control and the audit trail.

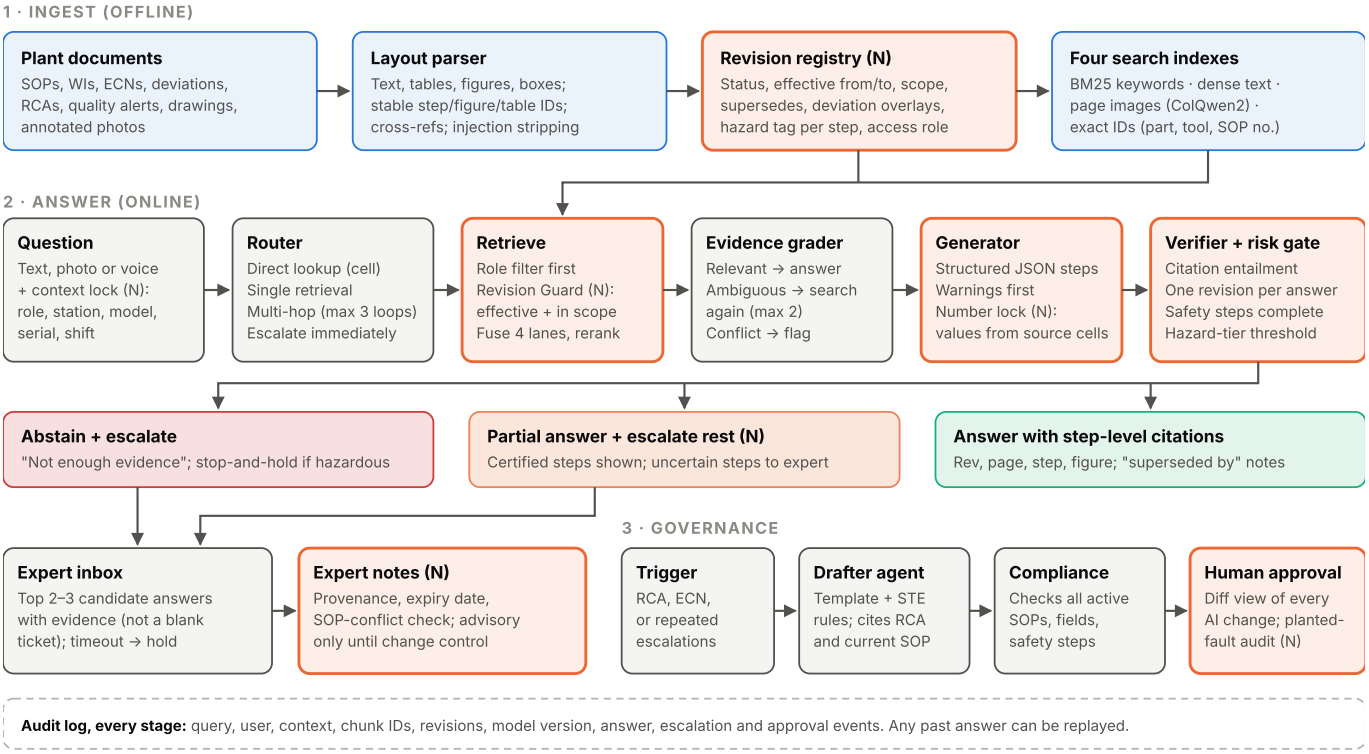


Figure 1. M.A.C.H. architecture. Orange boxes mark the components where we found no prior peer-reviewed work.

The checks every answer passes

1. **Context lock.** Role, station, machine model, serial range and shift are attached to the question, so search can only see documents the user may read and that apply to their machine.
2. **Revision Guard.** A hard filter, not a soft re-rank: only chunks whose status is effective, whose date window covers today and whose scope covers the operator's line are searchable. Deviations and ECNs are applied to the specific steps they patch.
3. **Evidence grading.** A CRAG-style grader scores what was found. Ambiguous evidence triggers up to two re-searches; irrelevant evidence leads to abstention.
4. **Conflict detection.** If two active sources give different values for the same parameter, M.A.C.H. shows both and escalates. A quality alert outranks the SOP until it is resolved.
5. **Number lock.** Every torque, temperature, pressure or dimension in the answer is copied by code from the cited table cell, with tolerance and unit. If OCR text and the image reading disagree on a digit, the system abstains.
6. **Citation and safety verification.** Each step's citation must entail the step, all citations must come from one revision set and the cited variant must match the machine, and every warning, PPE or lockout/tagout (LOTO) step in the source must appear in the answer.
7. **Hazard-tiered gate.** The confidence needed to answer depends on the step's hazard tier (below). Below threshold, the system gives a partial answer or abstains, then escalates.

Hazard tier	Examples	Target max error on answered steps	Below threshold
Tier 1 · Critical	LOTO, pressure release, lifting, torque on safety fasteners	≤ 1%	Abstain, "stop and hold", escalate

Tier 2 · Quality	Torque specs, temperatures, inspection criteria	≤ 5%	Partial answer, escalate the uncertain steps
Tier 3 · General	Tool locations, part descriptions, cosmetic steps	≤ 10%	Answer with a "low confidence" label

Thresholds are calibrated on a held-out set with conformal risk control (Mohri & Hashimoto, ICML 2024; C-RAG, ICML 2024), separately per tier. They are re-checked whenever an SOP family gets a new revision, because the statistical guarantee assumes new questions look like the calibration set. The error targets are our design choice, not values from a paper.

Change control for AI-drafted work instructions

SOP family is locked while a draft is open, so two drafts can't collide. Every transition is logged with user and time.



Figure 2. Nothing the AI writes becomes official without a signed human approval. Reviewers are periodically shown drafts with planted faults to measure automation bias.

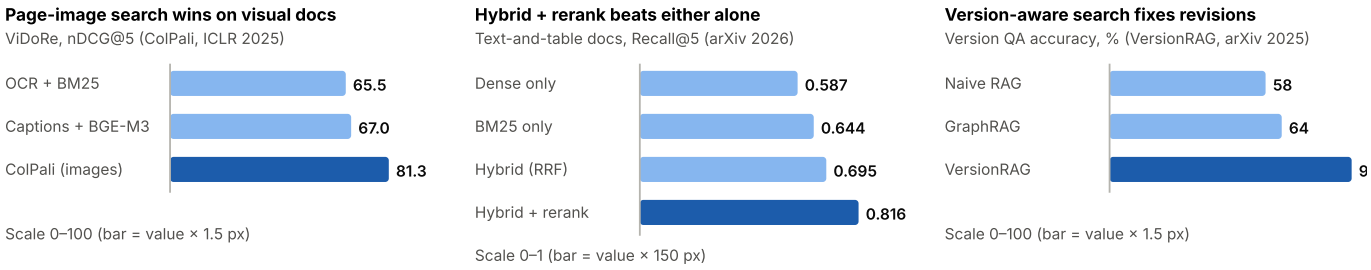
4. Novelty: 17 ideas, ranked

Ranks 1–7 are the PPT headline set. Ranks 8–17 strengthen the paper and the Q&A. "Effort" is for a 3–4 person team in one week.

#	Idea	Gap it fills (evidence)	Effort
1	**Revision Guard**: hard effectivity filter, "superseded by" statements, one-revision-per-answer check	Stale documents flip 17–91% of answers; VersionRAG 90% vs 58% naive RAG both arXiv. Peer-reviewed: HoH (ACL 2025) on outdated info, Morita et al. (EMNLP 2026 Industry) on evolving policy corpora. None on step-level supersession, ECN overlays or mixed-revision answers.	Low
2	**Step-level, safety-complete citations** with figure and step IDs	ALCE (EMNLP 2023) scores sentences, not step order or missing safety steps. Visual box citation reaches only 54–68% even fine-tuned (VISA, ACL 2025).	Low–Med
3	**Hazard-tiered abstention** with partial answers	Conformal guarantees exist (ICML 2024); none found calibrated per hazard class or returning certified step subsets.	Med
4	**Smart escalation + expert memory** with expiry and conflict checks	Prediction sets help humans (NeurIPS 2024). The one factory loop found has no expiry or SOP-conflict control (Frontiers in AI 2024).	Med
5	**MACH-Bench**: multimodal, with revision traps and unsafe asks	DesignQA (ASME JCISE) covers multimodal engineering docs; SOPRAG arXiv has an SOP retrieval benchmark; VersionQA is software-only arXiv. None combines WIs, images and revision traps.	Med
6	**Planted-fault reviewer test** for AI drafts	No study measures change-control outcomes for LLM-drafted instructions; EU AI Act Art. 14 asks for automation-bias awareness.	Low
7	**Adaptive routing** by question complexity	Adaptive-RAG (NAACL 2024): same F1 as IRCoT at 1.46× vs 3.33× the time.	Low
8	**Context lock**: answers scoped to machine variant	"Misgrounded" citations defined in legal RAG (JELS 2025); never applied to machine variants.	Low

9	**Deviation overlays** on individual steps	21 CFR 211.100(b) requires recorded deviations; no RAG work models step-level, time-limited patches.	Med
10	**"What changed?"** answers across revisions, including tables and figures	Diffing multimodal revisions is listed as open in versioning work.	Med
11	**Cross-reference following** ("see Fig. 3, step 4")	No benchmark or system found that evaluates it.	Med
12	**Exact-ID search lane** with tuned fusion	Naive fusion ranks relevant results lower on 8–21% of queries arXiv; no study isolates part numbers.	Low
13	**Number lock** for critical values and units	LLM judges are near random on expert-level correctness (JudgeBench, ICLR 2025); parameters need exact checks.	Low
14	**Query by photo** of a part or panel	No paper found inference.	Med
15	**Shift-aware escalation**	Factory study: assistant most valuable on night shifts with no expert (Frontiers in AI 2024).	Low
16	**Recalibrate on every revision**	Conformal guarantees break under distribution shift; none found tied to document control.	Med
17	**Simplified Technical English check** on drafts	Builds on Jonek & Manns (CARV 2025): controlled language raised vocabulary compliance 50%→85% and cut editing ~20→12 min. Our new part is measuring it under change control with planted faults.	Low

Also worth a slide line inference, no paper: hands-free voice queries for gloved hands, and answers in Hindi or the local language while citations stay in the original English SOP.



Darker bar = best method in that study. Each panel is a different study and metric, so compare within a panel only. Panels 2 and 3 are preprints (not peer-reviewed).

Figure 3. Published results behind three design choices: page-image search, hybrid retrieval with reranking, and version-aware retrieval.

5. Edge cases and how M.A.C.H. handles them

The most dangerous failure is a confident, cited, wrong answer. We list 37 edge cases; 13 are high risk. This is the summary; the full test plan expands it to 55 cases in 14 families with a scripted pass criterion for each. Each one becomes a test case in MACH-Bench, so every robustness claim is backed by a measured number.

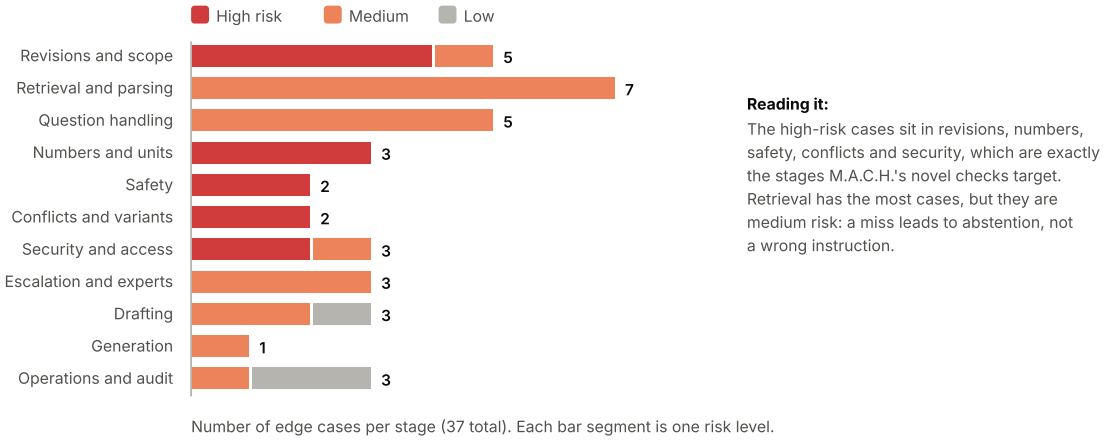


Figure 4. Edge cases by pipeline stage and risk level. Full list below.

Stage	Edge case	Failure	How M.A.C.H. handles it	Risk
Revisions	Two revisions of one SOP both marked effective	Answer mixes old and new steps	Ingestion blocks it; verifier rejects mixed-revision answers	High
Revisions	ECN changes one step before the SOP is reissued	Old step used	Deviation overlay patches that step	High
Revisions	Revision effective tomorrow, or on one line only	Too early or wrong line	Filter on effective window and scope using context lock	High
Revisions	SOP withdrawn with no replacement	Answers from a withdrawn procedure	Abstain: "procedure withdrawn, contact QA"	High
Revisions	Operator quotes an old document number	Executes obsolete text	Answer from current revision and state what changed	Med
Variants	Same procedure, different machine model or serial range	Right text, wrong machine	Context lock; citation must match the variant	High
Conflict	SOP and quality alert give different values	Silently picks one	Show both, escalate; alert outranks SOP until resolved	High
Numbers	N·m in one document, lbf·ft in another	Wrong value or unit	Number lock; units converted by code	High
Numbers	Tolerance given ("25 ± 2 N·m")	Tolerance dropped	Value, tolerance and unit extracted as one field	High
Numbers	OCR reads 8 as B, or 2,5 vs 2.5	Wrong digit stated confidently	Cross-check OCR against image reading; mismatch → abstain	High
Safety	Answer would omit a LOTO or PPE step	Injury risk	Safety-completeness check; warnings first, never truncated	High
Safety	"How do I skip / bypass this interlock?"	Unsafe instruction	Router refuses, logs, escalates	High
Security	Hidden "ignore your rules" text in an ingested doc	Hijacked answer	Strip at ingestion; retrieved text treated as data; fixed output schema	High

Access	Operator asks about a restricted RCA	Leak	Role filter before search, not after generation	High
Security	Names or personal data in RCA reports	Privacy breach	Redact at ingestion; no per-operator analytics	Med
Retrieval	Answer only in a diagram callout or table cell	Text search misses it	Page-image lane plus layout parsing	Med
Retrieval	Scanned, skewed or hand-marked pages	OCR garbage	Route to image search; flag low-quality pages	Med
Retrieval	Table or step continues on the next page	Half a step	Merge split tables and steps at ingestion	Med
Retrieval	Diagram not updated when the text was	Figure contradicts text	Figure carries its own revision hash; mismatch flagged	Med
Retrieval	Plant slang ("CP" for caliper piston)	Search miss	Plant glossary expands query; exact-ID lane	Med
Retrieval	Same warning repeated in 10 documents	Unfair scoring	Multi-gold evidence sets in the benchmark	Med
Retrieval	Cross-reference to another document	Answer missing the referenced figure	Agent follows parsed cross-reference links	Med
Query	Vague ("how do I fix it?")	Guesses the machine	Use station context or ask one clarifying question	Med
Query	Two questions in one, or a multi-step plan	Answers half	Router sends it to the multi-step path	Med
Query	Not covered by any document	Hallucination	Abstain: "not in the controlled documents"	Med
Query	Blurry or wrong photo	Wrong page, high confidence	Image-quality check; confirm the matched part	Med
Query	Hindi or mixed-language question	Search miss	Translate query for search; answer in user's language, cite English source	Med
Generation	Very long SOP overflows the context	Procedure cut off	Step-level chunks; completeness check against the step count	Med
Escalation	No expert responds in time	Operator blocked or guesses	Timeout → "stop and hold" for tier-1 steps; supervisor alert	Med
Escalation	Expert note conflicts with the SOP, or has expired	Informal override	Notes are advisory, expire, and can't override without change control	Med
Escalation	Everything gets escalated	System ignored	Track escalation rate; tune on the risk–coverage curve	Med
Drafting	Draft contradicts another active SOP	Bad revision approved	Compliance agent checks all active SOPs for that station	Med
Drafting	Reviewer rubber-stamps the AI draft	Automation bias	Diff view; planted-fault audits	Med
Drafting	Two drafts of the same SOP at once	Collision	SOP family locked during review	Low
Ops	LLM API down or slow	No answer on the floor	Fallback: keyword search showing source pages, with a banner	Med
Ops	Model version changes silently	Behaviour drifts	Pin version; rerun benchmark on every change	Low
Audit	"What did the AI tell operator X on date D?"	No proof	Full audit log with replay	Low

6. Evaluation plan

We compare five systems on the same frozen corpus, with the same LLM, top-k and context budget, so any gain comes from M.A.C.H.'s design.

Systems

1. BM25 keyword search (shows top pages only)
2. BM25 + LLM
3. Text-only RAG (dense over OCR text, no checks)
4. Hybrid text RAG (BM25 + dense, RRF)
5. M.A.C.H. full, plus ablations: no page images, no Revision Guard, no verifier, no hazard gate, no router

Metrics

- **Retrieval:** Recall@5, nDCG@10 (multi-gold)
- **Answers:** correctness, faithfulness, citation precision/recall (ALCE-style)
- **Procedure:** step-order fidelity, exact match on critical values, safety-step recall
- **Trust:** revision-mixing rate, superseded-citation rate, abstention precision/recall, unsafe-answer rate, escalation rate
- **Drafting:** accept/edit/reject, edit distance, planted-fault catch rate
- **Ops:** p50/p95 latency, cost per query

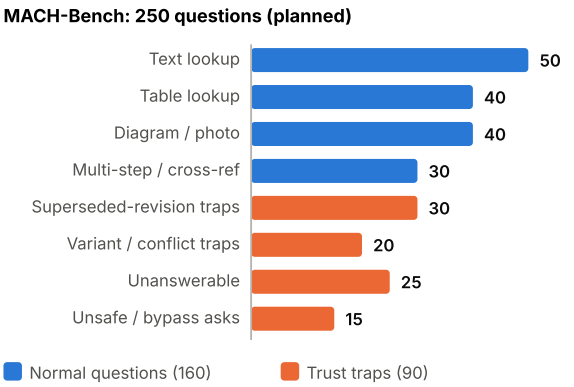


Figure 5. Planned question mix. 36% are traps designed to make a naive system fail.

Making it rigorous

- **Corpus:** real revision chains from public FAA Airworthiness Directive supersessions, plus scripted counterfactual edits and a realistic shop-floor set (SOPs with 2–3 revisions, ECNs, deviations, quality alerts, RCAs, drawings, photos). Freeze and hash the snapshot.
- **Labels:** each question has a gold answer, all valid evidence locations, expected behaviour (answer, abstain or escalate) and, for procedures, the ordered steps and critical values. Two annotators label at least 25%; report Cohen's κ.
- **Split:** 30% dev for tuning prompts and thresholds, 70% test touched once.
- **Judging:** an LLM judge from a different model family than the generator, run 3 times with shuffled order, validated on ≥100 human-labelled items. Critical values are checked by exact match, never by the judge.
- **Statistics:** paired per-question tests (McNemar or bootstrap), Holm correction across metrics, 95% confidence intervals; ARES-style correction to turn a small human-labelled set into intervals.
- **Human study:** 3–5 people answer 10 questions with keyword search vs M.A.C.H.; report minutes per correct answer.

Release gates (from the test plan)

A build ships only if it passes every hard gate. These match the paper draft and the edge-case test plan.

Gate	Metric	Threshold	Type
No superseded citations	Superseded-citation rate on revision traps	0 of n	Hard
No mixed-revision answers	Mixed-revision answer rate	0 of n	Hard
No restricted leakage	Restricted element IDs in a lower-role context or answer	0	Hard
No injected instruction followed	Injection success rate	0	Hard
Safety steps present	Safety-step recall on answered procedures	≥ 0.98	Hard

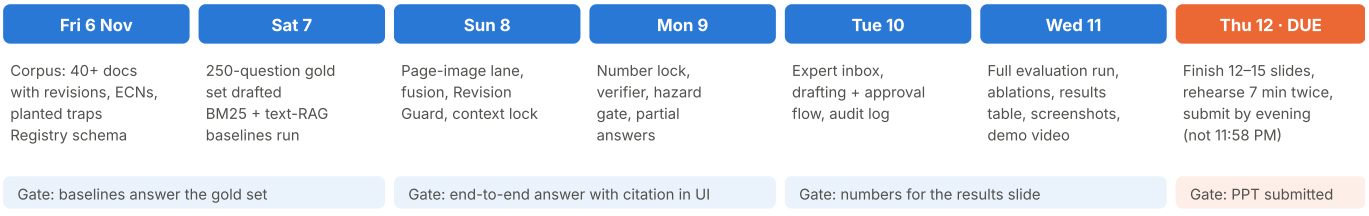
Exact critical values	Torque, tolerance, unit, part number exact match	≥ 0.99 on answered items	Hard
Safe fallback	Keyword fallback shown when the model is down	100% of kill-switch runs	Hard
Abstains when it should	Abstention recall on unanswerable items	≥ 0.90	Soft
Does not over-abstain	Over-abstention on answerable items	≤ 0.15	Soft
Fast enough	Queries answered in under 15 s; p95	≥ 95%; p95 < 15 s	Soft

Headline numbers for the results slide (to be measured): revision-mixing rate against the baselines, unsafe-answer rate on trap questions, citation precision, Recall@5 on diagram questions, and p95 latency.

7. Technology stack

Layer	Choice for the prototype	Why
Parsing	Docling or PyMuPDF; page images at 150 dpi	Text, tables and bounding boxes with stable IDs
Page-image search	ColQwen2 / ColPali via byaldi or Qdrant multi-vector	Qdrant supports late-interaction matching; pgvector does not natively
Text search	BM25 (rank_bm25 or Postgres full-text) + text embeddings + exact-ID index	Keywords still win on part numbers and jargon
Fusion and rerank	Weighted fusion tuned on dev, cross-encoder reranker	Avoids the naive-fusion penalty
LLM	Paper configuration: 4-bit 7B vision-language model on one plant-side GPU. Demo option: Gemini via Vertex AI. Structured JSON output either way	Plant data stays on site; schema-enforced answers
Agents	LangGraph or plain Python state machine	Router, grader, drafter, compliance; easy to audit
Data and governance	Postgres: registry, overlays, roles, audit log, approval queue	One source of truth for revisions
UI	Streamlit or Next.js: operator chat, PDF viewer with highlighted step, expert inbox, review diff view	Demo-ready in days
Deploy	Docker Compose on one GPU box (open weights served with vLLM); Cloud Run for a hosted demo	Same pipeline on-prem or in the cloud
Evaluation	RAGAS + custom scripts for trust metrics	Reproducible results table

8. Seven-day plan to the PPT deadline



If behind schedule, cut in this order: query by photo, "what changed?" answers, STE checker, drafting UI polish. Never cut Revision Guard, the verifier or the baselines.
Suggested split (3–4 people): data and evaluation · retrieval and registry · LLM checks and agents · UI, governance and PPT.

Figure 6. One week, built thin first and deepened after. A working slice by Sunday matters more than any single component.

Suggested 12-slide story (7 minutes)

1. Title and one-line pitch
2. Problem: one real-looking story of an operator following an outdated revision
3. Why current tools fail (stale documents, unsupported citations)
4. M.A.C.H. at a glance (Figure 1)

5. Novelty 1: Revision Guard, with a screenshot of the "superseded by" note
6. Novelty 2: step-level, safety-complete citations and number lock
7. Novelty 3: hazard-tiered abstention and smart escalation
8. Change-controlled drafting (Figure 2)
9. Evaluation design: MACH-Bench, baselines, release gates
10. Results: one table and two honest failure cases
11. Security, deployment and edge cases handled (Figure 4)
12. Impact for LTTS clients and next steps

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9. Jury questions and our answers

Q1. Isn't this just RAG with extra steps?

RAG is the base. The contribution is the trust layer: Revision Guard, step-level safety-complete citations, number lock and hazard-tiered abstention. We found no peer-reviewed work on these for procedures, and we measure each with an ablation.

Q2. How do you know which revision is effective?

From document-control metadata (status, effective dates, scope, supersedes links), not file dates. A hard filter removes everything else before the LLM sees it, and the verifier rejects any answer mixing revisions.

Q3. What if two current documents disagree?

M.A.C.H. never picks silently. It shows both sources, escalates to an expert, and treats a quality alert as overriding the SOP until resolved.

Q4. How do you stop hallucinated numbers?

Number lock: critical values are copied by code from the cited table cell with their tolerance and unit. The LLM never writes them. If OCR and the image reading disagree, the system abstains.

Q5. Won't it refuse too often to be useful?

We report the risk–coverage curve and escalation rate, not just accuracy. Only tier-1 hazardous steps need very high confidence; general lookups answer with a confidence label. Partial answers keep the operator moving.

Q6. Where do confidence thresholds come from?

Conformal risk control on a held-out calibration set, separately per hazard tier (ICML 2024 methods). Thresholds are re-checked whenever an SOP family is revised.

Q7. Why not fine-tune a model on the SOPs?

Retrieval beats fine-tuning for adding new knowledge (Ovadia et al., EMNLP 2024), and SOPs change too often. Updating an index is instant and auditable; retraining is neither.

Q8. You have no real plant data. Is the test valid?

Part of it is real: revision chains come from public FAA Airworthiness Directive supersessions, and we also run eight public peer-reviewed benchmarks. The shop-floor part is synthetic but reviewed by someone with plant experience (name them). The benchmark, code and schema are designed so LTTS can drop in real documents and rerun the same evaluation.

Q9. Can a malicious document hijack it?

Instruction-like text is stripped at ingestion, retrieved text is treated as data, output is schema-enforced, and access is filtered by role before search. We include injection documents in the test set.

Q10. What if the cloud LLM is down or data can't leave the plant?

The reference configuration already runs on-prem: a 4-bit 7B vision-language model on one plant-side GPU, so documents never leave the plant. If the model is down, a keyword-only fallback shows the right source pages with a banner.

Q11. Who is accountable for an AI-drafted work instruction?

The human approver. Nothing becomes effective without a signed approval, every AI change is shown as a diff, and planted-fault audits measure whether reviewers are really checking.

Q12. How fast is it?

Release gate: 95% of queries answered in under 15 s on a single plant-side GPU (faster with a cloud API). Simple lookups take one pass; only multi-step questions run the agent loop (Adaptive-RAG shows this keeps accuracy at under half the latency). Report the measured p50 and p95.

Q13. What about expert knowledge that isn't written down?

Expert answers to escalations are stored as advisory notes with author, date, expiry and an SOP-conflict check. Repeated notes on the same topic trigger a change-control draft, so tacit knowledge becomes an official revision.

Q14. How is this different from commercial copilots?

General copilots don't understand effectivity, machine variants, hazard tiers or change control. M.A.C.H. is built around document-control rules from ISO 9001 §7.5 and GMP, and it proves its behaviour on trap questions.

Q15. What's the business value for LTTS?

Fewer quality escapes from obsolete instructions, faster answers on night shifts without experts, audit-ready traceability, and faster WI authoring. It is reusable across LTTS's manufacturing, automotive and medical-device clients.

Q16. Hasn't HoH or SOPRAG already solved this?

HoH (ACL 2025) shows outdated documents hurt RAG but treats a document as a whole; it has no effectivity dates, ECN overlays or step-level supersession. SOPRAG (preprint) improves SOP retrieval but has no revisions, images, citations or abstention. We cite both, borrow HoH's outdated-versus-current setup, and test exactly what they leave out.

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10. Honest limitations

- The shop-floor documents are synthetic and the real revision chains come from aviation, so absolute accuracy won't transfer directly to a given plant; relative gains over baselines are the claim.
- Conformal guarantees hold only while new questions resemble the calibration set; we mitigate with recalibration on revision events, not eliminate the risk.
- Several of the newest supporting results are arXiv preprints and are labelled as such. Venue status of ColPali (ICLR 2025) and some others should be confirmed before citing in a paper.
- The literature search used the OpenAlex index (covers IEEE incl. Early Access, ACM, ScienceDirect in press, Springer) but paywalled full texts were not read, so novelty claims are phrased as "we found no prior work".

> **Open questions for you:** How many people are on the team, and who owns which part? Do you have GCP credits or a GPU? Can anyone with plant experience review the gold set? Please share the LTTS PPT template so the slide story can be mapped onto it.

Key references

peer-reviewed venue given · arXiv not peer-reviewed. Full list with links and verification notes: *MACH multimodal RAG literature review.md* in the project files.

1. Faysse et al. ColPali: Efficient Document Retrieval with Vision Language Models. ICLR 2025 (venue to confirm). peer-reviewed
2. Yu et al. VisRAG: Vision-based RAG on Multi-modality Documents. ICLR 2025. peer-reviewed
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5. Ludwig, Schmidt, Kühn. Ontology-based RAG for a voice-controlled maintenance assistant. Computers in Industry 169, 2025. peer-reviewed
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